

THREE SQUARES IN A RECTANGLE

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ABSTRACT. For $x \geq 1$, let $G_3(x)$ be the maximum sum of the side lengths of three pairwise interior-disjoint, arbitrarily rotated squares contained in a $1 \times x$ rectangle. We determine this function exactly: $G_3(x) = x + \frac{1}{2}$ for $1 \leq x \leq \frac{3}{2}$, $G_3(x) = 2$ for $\frac{3}{2} \leq x \leq 2$, $G_3(x) = x$ for $2 \leq x \leq 3$, and $G_3(x) = 3$ for $x \geq 3$. This completes the $n = 3$ case of the rectangular square-packing question posed by Richard Stanley in a 2021 MathOverflow comment. The values for $\frac{3}{2} \leq x \leq 3$ follow from a strip theorem stating that three squares in $[0, 1] \times [0, H]$, $H \geq 2$, have total side length at most H . For $x \geq 3$, the formula is immediate because each square has side length at most 1. The range $1 \leq x \leq \frac{3}{2}$ is handled by combining the two-square theorem with an additional semi-perimeter estimate for a triangle whose two nonhorizontal sides have opposite slopes. In particular, the special case $x = 2$ answers the question asked in the MathOverflow post. In connection with Erdős problem #106, we also prove that every five-square packing in the unit square with an axis-parallel guillotine cut has total side length at most 2.

1. STATEMENT AND CONTEXT

For a positive integer n , let $f(n)$ denote the maximum total side length of n interior-disjoint squares, of arbitrary orientation, packed into a unit square. Erdős conjectured that $f(k^2 + 1) = k$; this is problem #106 on Bloom's index of Erdős problems [4]. Erdős and Soifer [3] and, independently, Campbell and Staton [2] established the lower bound $f(k^2 + 2c + 1) \geq k + c/k$ for integers c with $|c| < k$, and conjectured that equality holds; Praton [6] showed this generalized conjecture to be equivalent to Erdős's original conjecture, and Singh [7] showed that the original conjecture is equivalent to the convergence of $\sum_{k \geq 1} (f(k^2 + 1) - k)$. Staton and Tyler [8] introduced the axis-parallel variant $g(n)$, for which Baek, Koizumi and Ueoro [1] proved the conjectured formula. The present note concerns a rectangular variant in which rotations are unrestricted. In a 2021 comment on a MathOverflow question, Richard Stanley asked more generally for the maximum sum of the side lengths of n squares in a $1 \times x$ rectangle, for any real $x > 1$ [5]. We determine this quantity exactly for $n = 3$ and every $x \geq 1$; see Theorem 9. The main geometric ingredient is the following strip theorem.

Theorem 1 (Strip Theorem). *Let $H \geq 2$ and let three squares with pairwise disjoint interiors lie inside the rectangle $R = [0, 1] \times [0, H]$. The squares may be arbitrarily rotated. Then the sum of their side lengths is at most H .*

For $H \geq 3$ this is immediate, since each side is at most 1 (see Step 1 below), so the content of the theorem is the range $2 \leq H \leq 3$. The case $H = 2$ is exactly the question asked on MathOverflow in July 2021 [5]: three interior-disjoint squares in a 1×2 rectangle have total side length at most 2.

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Relation to the existing partial answers. A partial answer to the MathOverflow question introduces two separating lines and bounds each square by the largest square inscribed in a triangle using Pólya's formula, but its monotonicity analysis is completed only under an unproven auxiliary assumption $a, d < 3.66$ on the line parameters [5]. The proof below keeps the same separating-line setup and replaces the inscribed-square analysis by elementary semi-perimeter lemmas for enclosing rectangles (Section 2). This removes all restrictions on the separating-line slopes and yields the full $1 \times H$ statement.

Sharpness. The threshold $H = 2$ is sharp: for every $1 \leq H < 2$, the constructions in Remark 7 have total side length greater than H .

Connection with Erdős problem #106. Corollary 11 shows that every five-square packing in the unit square with an axis-parallel guillotine cut has total side length at most 2.

Notation. A square of side s with center (x, y) and orientation θ has perpendicular edge directions $u = (\cos \theta, \sin \theta)$ and $v = (-\sin \theta, \cos \theta)$. Write

$$\rho = |\cos \theta| + |\sin \theta| \in [1, \sqrt{2}], \quad w = s\rho.$$

Here w is the side of the axis-aligned bounding box (both horizontal and vertical sides equal w), and $\rho \in [1, \sqrt{2}]$ measures how much wider than s the square becomes when rotated. For $\alpha \in \mathbb{R}$ set

$$\psi(\alpha) = |\alpha \cos \theta + \sin \theta| + |-\alpha \sin \theta + \cos \theta|,$$

so that the width of the square in the unit direction $(\alpha, 1)/\sqrt{1 + \alpha^2}$ equals $s\psi(\alpha)/\sqrt{1 + \alpha^2}$, and its half-width is $s\psi(\alpha)/(2\sqrt{1 + \alpha^2})$ (the support function of a square); note $\psi(0) = \rho$.

Throughout, the *semi-perimeter* of a $p \times q$ rectangle is $p + q$. We use repeatedly the elementary identity

$$\text{span}\{0, p, p + q\} = \frac{1}{2}(|p| + |q| + |p + q|), \quad (1.1)$$

where span of a finite set of reals is its maximum minus its minimum, together with the fact that if $p + q + r = 0$ then $p_+ + q_+ + r_+ = \frac{1}{2}(|p| + |q| + |r|)$, where $x_+ = \max(x, 0)$ denotes the positive part.

2. THREE SEMI-PERIMETER LEMMAS

Fact 2 (flush enclosing rectangle). Among all rectangles, of any orientation, containing a given convex polygon, one of minimum perimeter has a side collinear with an edge of the polygon.

Proof. This is the perimeter counterpart of the minimum-area statement obtained by rotating calipers; compare [9]. For an angle φ , let $\sigma(\varphi)$ be the semi-perimeter of the bounding box of the polygon in the frame rotated by φ . Every enclosing rectangle at angle φ contains that bounding box, hence has semi-perimeter at least $\sigma(\varphi)$; it therefore suffices to locate the minimum of σ , which exists since σ is continuous and $\pi/2$ -periodic. By definition $\sigma(\varphi)$ is the sum of the lengths of the orthogonal projections of the polygon onto $e_1(\varphi) = (\cos \varphi, \sin \varphi)$ and $e_2(\varphi) = (-\sin \varphi, \cos \varphi)$. Each such length equals $(z - z') \cdot e_j(\varphi)$ for the pair z, z' of vertices supporting the polygon in that direction; hence on any arc of angles on which these supporting vertices do not change, both lengths are linear in $(\cos \varphi, \sin \varphi)$, so $\sigma(\varphi) = A \cos \varphi + B \sin \varphi$ and $\sigma'' = -\sigma < 0$ there. Thus σ is strictly concave on such an arc and any interior critical point is a strict local maximum; the minimum is therefore attained at an angle at which a supporting vertex changes. At such an angle a supporting line contains an edge of the polygon, so the corresponding side of the bounding box is collinear with that edge. $\square$

Lemma 3. *Every rectangle containing a right triangle with legs $\lambda, \mu \geq 0$ has semi-perimeter at least $\lambda + \mu$.*

Proof. If $\lambda\mu = 0$ the triangle degenerates to a segment of length $\lambda + \mu$, and any rectangle containing it has semi-perimeter at least its diagonal, hence at least $\lambda + \mu$; assume therefore $\lambda, \mu > 0$. By Fact 2 it suffices to check the orientations flush with an edge. Flush with either leg, the bounding box is $\lambda \times \mu$, of semi-perimeter $\lambda + \mu$. Flush with the hypotenuse, of length $r = \sqrt{\lambda^2 + \mu^2}$, the box is $r \times \frac{\lambda\mu}{r}$ (base times altitude), and

$$r + \frac{\lambda\mu}{r} - (\lambda + \mu) = \frac{r^2 + \lambda\mu - r\lambda - r\mu}{r} = \frac{(r - \lambda)(r - \mu)}{r} \geq 0,$$

since $r \geq \max(\lambda, \mu)$. $\square$

Lemma 4. *For $a, b \in \mathbb{R}$ let $T(a, b) = \text{conv}\{(0, 0), (a, 1), (b, 1)\}$ and let $C = \max(0, a, b) - \min(0, a, b)$ be its horizontal span, so that $C = \frac{1}{2}(|a| + |a - b| + |b|)$ by (1.1). Then every rectangle containing $T(a, b)$ has semi-perimeter at least*

$$1 + C - |a|, \quad \text{and, symmetrically,} \quad 1 + C - |b|.$$

Proof. We prove the first bound; the second follows by symmetry, swapping a and b . Set $L = C - |a|$. One checks directly that L is the distance from b to the interval $[\min(0, a), \max(0, a)]$, so $L \geq 0$.

Case 1: $L = 0$. Here b lies in the interval $[\min(0, a), \max(0, a)]$, so the bound reduces to $1 + C - |a| = 1 + L = 1$ and a crude estimate suffices. The triangle has one vertex at height 0 and another at height 1, so it contains two points whose distance is at least 1. Any $p \times q$ rectangle containing these two points has diagonal at least their distance, so $\sqrt{p^2 + q^2} \geq 1$. Since $p + q \geq \sqrt{p^2 + q^2}$, we conclude $p + q \geq 1 = 1 + L$.

Case 2: b on the opposite side of a . By the reflection $x \mapsto -x$ if needed we may assume $b < 0 \leq a$, so $L = |b|$. Since $b < 0 \leq a$, the point $(0, 1)$ lies on the top edge between $(b, 1)$ and $(a, 1)$, so $T(a, b)$ contains the right triangle $\Delta = \text{conv}\{(0, 0), (0, 1), (b, 1)\}$, whose legs are 1 and L . By Lemma 3, every rectangle containing Δ has semi-perimeter at least $1 + L$; since $T(a, b) \supseteq \Delta$, the same bound holds for every rectangle containing $T(a, b)$.

Case 3: b on the same side as a . After a reflection we may assume $0 \leq a < b$, so $L = b - a$. Write $O = (0, 0)$, $P = (a, 1)$, $Q = (b, 1)$. By Fact 2 we check the three flush orientations.

Flush with OP . Let $\ell = \sqrt{a^2 + 1} = |OP|$, and project onto the unit vector $(a, 1)/\ell$ along OP . The images of $O = (0, 0)$, $P = (a, 1)$, $Q = (b, 1)$ are

$$0, \quad \frac{a^2 + 1}{\ell} = \ell, \quad \frac{ab + 1}{\ell},$$

respectively. Since $0 \leq a < b$ we have $ab \geq a^2$, hence $\frac{ab+1}{\ell} \geq \ell$, so the images run from 0 up to $\frac{ab+1}{\ell}$. Using $ab + 1 = (a^2 + 1) + a(b - a) = \ell^2 + aL$, the span along OP is

$$\frac{ab + 1}{\ell} = \frac{\ell^2 + aL}{\ell} = \ell + \frac{aL}{\ell}.$$

Perpendicular to OP , projecting onto $(-1, a)/\ell$, the images of O, P, Q are 0, 0, $-L/\ell$; the span (maximum minus minimum) is therefore L/ℓ . Hence the semi-perimeter is

$$\ell + \frac{aL}{\ell} + \frac{L}{\ell} = \ell + \frac{L(a + 1)}{\ell} \geq 1 + L,$$

because $\ell \geq 1$ and $a + 1 \geq \sqrt{a^2 + 1} = \ell$.

Flush with OQ . Let $m = \sqrt{b^2 + 1} = |OQ|$, and project onto the unit vector $(b, 1)/m$ along OQ . The images of O, P, Q are $0, (ab + 1)/m, m$, with $0 \leq (ab + 1)/m \leq m$ (the upper bound since $ab + 1 \leq b^2 + 1 = m^2$, using $a \leq b$), so the span along OQ is m . Perpendicular to OQ , projecting onto $(-1, b)/m$, the images of O, P, Q are $0, L/m, 0$, so that span is L/m . Hence the semi-perimeter is $m + \frac{L}{m}$. Since $m = \sqrt{b^2 + 1} \geq 1$ and $m \geq b \geq L$, both factors in

$$m + \frac{L}{m} - (1 + L) = \frac{m^2 + L - m - mL}{m} = \frac{(m - 1)(m - L)}{m}$$

are nonnegative, so $m + \frac{L}{m} \geq 1 + L$.

Flush with PQ (the horizontal top edge). The bounding box is $[0, b] \times [0, 1]$, of semi-perimeter $b + 1 \geq L + 1$, since $b \geq b - a = L$. $\square$

Lemma 5. *Let $p, q \geq 1$ and let*

$$T_{p,q} = \text{conv}\{(0, 0), (-p, 1), (q, 1)\}.$$

Every rectangle containing $T_{p,q}$ has semi-perimeter at least

$$p + q + \frac{1}{2}.$$

Proof. Put $C = p + q$. By Fact 2, it is enough to check the three orientations flush with an edge of the triangle.

Flush with the horizontal edge, the bounding box has semi-perimeter $C + 1$. For the edge from $(0, 0)$ to $(-p, 1)$, put $r = \sqrt{p^2 + 1}$. Projection onto $(-p, 1)/r$ gives the three values

$$0, \quad r, \quad \frac{1 - pq}{r},$$

and projection onto the perpendicular unit vector $(1, p)/r$ gives $0, 0, C/r$. Since $pq \geq 1$, the two projection spans are pC/r and C/r ; thus the semi-perimeter of this bounding box is

$$\frac{C(p + 1)}{r}.$$

Now $C \geq p + 1 \geq r$, while

$$r - p = \frac{1}{r + p} \leq \frac{1}{2},$$

so

$$\frac{C(p + 1)}{r} - C = \frac{C}{r}(p + 1 - r) \geq \frac{1}{2}.$$

The orientation flush with the edge from $(0, 0)$ to $(q, 1)$ is symmetric. Hence every flush bounding box, and therefore every enclosing rectangle, has semi-perimeter at least $C + \frac{1}{2}$. $\square$

3. PROOF OF THE STRIP THEOREM

Let the squares be A, B, C , with sides s_A, s_B, s_C , orientations $\theta_A, \theta_B, \theta_C$ and centers $c_i = (x_i, y_i)$, and suppose for contradiction that

$$S := s_A + s_B + s_C > H.$$

Choose $\varepsilon > 0$ with $(1 - \varepsilon)S > H$, shrink each square by the factor $1 - \varepsilon$ about its center, and relabel; all the separations below are then strict, so we may argue with strict disjointness and no touching degeneracies.

Step 1: a common vertical line. The horizontal projection of a square has length $w = s\rho \geq s$; since it must fit in $[0, 1]$, every side satisfies $s_i \leq 1$. In particular $S \leq 3$, so for $H \geq 3$ we are already done; assume from now on that $2 \leq H < 3$. For each pair,

$$s_i + s_j = S - s_k > H - 1 \geq 1.$$

If two squares had horizontal projections with disjoint interiors, those projections would occupy total length $s_i\rho_i + s_j\rho_j \leq 1$, forcing $s_i + s_j \leq 1$, a contradiction. So the three horizontal projections pairwise overlap; intervals have the Helly property in dimension one, hence there is a common abscissa x_0 : the vertical line $x = x_0$ meets all three squares. Their intersections with this line are disjoint segments; order the squares along it as A (bottom), B (middle), C (top).

Step 2: two separating lines. Disjoint compact convex sets can be separated by a line. A line separating A from B cannot be vertical, since their horizontal projections overlap in an open set, so its normal can be scaled to $(\alpha_1, 1)$; evaluating on the common vertical line shows that A lies on the side where $\alpha_1 x + y$ is smaller. Likewise a line with normal $(\alpha_2, 1)$ separates B (below) from C (above).

Consider the pair A, B . Projecting onto the unit normal $n_1 = (\alpha_1, 1)/\sqrt{1 + \alpha_1^2}$, the two squares occupy disjoint intervals, so the gap between their centers is at least the sum of their half-widths in this direction:

$$n_1 \cdot (c_B - c_A) \geq \frac{s_A \psi_A(\alpha_1) + s_B \psi_B(\alpha_1)}{2\sqrt{1 + \alpha_1^2}},$$

the right-hand side being the sum of the half-widths defined in the Notation. Since $n_1 \cdot (c_B - c_A) = (\alpha_1(x_B - x_A) + (y_B - y_A))/\sqrt{1 + \alpha_1^2}$, multiplying through by $\sqrt{1 + \alpha_1^2}$ gives

$$y_B - y_A \geq \frac{s_A}{2}\psi_A(\alpha_1) + \frac{s_B}{2}\psi_B(\alpha_1) - \alpha_1(x_B - x_A), \quad (3.1)$$

and the same argument applied to B, C with normal $n_2 = (\alpha_2, 1)/\sqrt{1 + \alpha_2^2}$ gives

$$y_C - y_B \geq \frac{s_B}{2}\psi_B(\alpha_2) + \frac{s_C}{2}\psi_C(\alpha_2) - \alpha_2(x_C - x_B). \quad (3.2)$$

Step 3: the master inequality. Adding (3.1) and (3.2), the left-hand sides telescope to $y_C - y_A$:

$$y_C - y_A \geq \frac{s_A}{2}\psi_A(\alpha_1) + \frac{s_B}{2}(\psi_B(\alpha_1) + \psi_B(\alpha_2)) + \frac{s_C}{2}\psi_C(\alpha_2) - X, \quad (3.3)$$

where

$$X := \alpha_1(x_B - x_A) + \alpha_2(x_C - x_B) = -\alpha_1 x_A + (\alpha_1 - \alpha_2)x_B + \alpha_2 x_C.$$

Since each square lies in R , its center keeps half a bounding box away from every side: $\frac{w_i}{2} \leq x_i \leq 1 - \frac{w_i}{2}$ and $\frac{w_i}{2} \leq y_i \leq H - \frac{w_i}{2}$. The vertical bounds give $y_C - y_A \leq H - \frac{1}{2}(w_A + w_C)$, so (3.3) becomes

$$\frac{s_A}{2}\psi_A(\alpha_1) + \frac{s_B}{2}(\psi_B(\alpha_1) + \psi_B(\alpha_2)) + \frac{s_C}{2}\psi_C(\alpha_2) + \frac{1}{2}(w_A + w_C) \leq H + X. \quad (3.4)$$

For X we use the horizontal bounds. Each center satisfies $\frac{w_i}{2} \leq x_i \leq 1 - \frac{w_i}{2}$, so for any $\gamma \in \mathbb{R}$ the linear form γx_i attains its maximum at one of the two endpoints, and in either case

$$\gamma x_i \leq \gamma_+ - \frac{|\gamma|w_i}{2}.$$

We apply this to the three terms of X , namely to $-\alpha_1 x_A$, to $(\alpha_1 - \alpha_2)x_B$ and to $\alpha_2 x_C$. Their coefficients $-\alpha_1, \alpha_1 - \alpha_2, \alpha_2$ sum to zero, so by the identity in the Notation the sum of their

positive parts equals $\frac{1}{2}(|\alpha_1| + |\alpha_1 - \alpha_2| + |\alpha_2|)$; write C_α for this quantity. Adding the three bounds therefore gives

$$X \leq C_\alpha - \frac{1}{2}(|\alpha_1|w_A + |\alpha_1 - \alpha_2|w_B + |\alpha_2|w_C). \quad (3.5)$$

Substituting (3.5) into (3.4) and moving every w_i term to the left-hand side, the coefficient of s_i there becomes, after $w_i = s_i \rho_i$,

$$\begin{aligned} K_A &= \frac{1}{2}(\psi_A(\alpha_1) + \rho_A(1 + |\alpha_1|)), \\ K_B &= \frac{1}{2}(\psi_B(\alpha_1) + \psi_B(\alpha_2) + \rho_B|\alpha_1 - \alpha_2|), \\ K_C &= \frac{1}{2}(\psi_C(\alpha_2) + \rho_C(1 + |\alpha_2|)), \end{aligned}$$

and we obtain the *master inequality*

$$K_A s_A + K_B s_B + K_C s_C \leq H + C_\alpha. \quad (3.6)$$

Step 4: the coefficients are semi-perimeters. At this point the coefficients K_i are purely algebraic, and nothing in their definition suggests a lower bound. The observation that drives the proof is that each of them is *exactly* the semi-perimeter of a bounding box of a triangle built from α_1, α_2 , read in the frame of the corresponding square; Lemmas 3 and 4 then bound them from below.

For a bounded set $V \subset \mathbb{R}^2$, the bounding box of V in the frame of square i , with axes u_i, v_i , has semi-perimeter $\text{span}(u_i \cdot V) + \text{span}(v_i \cdot V)$. Applying (1.1) to each of the two spans, one finds:

- K_A is the semi-perimeter of the bounding box, in the frame of A , of the right triangle $\Delta_{\alpha_1} = \text{conv}\{(0,0), (\alpha_1,0), (\alpha_1,1)\}$, whose legs are $|\alpha_1|$ and 1;
- K_B is the semi-perimeter of the bounding box, in the frame of B , of the triangle $T(\alpha_1, \alpha_2)$ of Lemma 4, whose horizontal span is exactly C_α ;
- K_C is the semi-perimeter of the bounding box, in the frame of C , of Δ_{α_2} , whose legs are $|\alpha_2|$ and 1.

We carry out the computation for K_A ; the other two are of the same shape. The vertices of Δ_{α_1} have u_A -images $0, \alpha_1 \cos \theta_A, \alpha_1 \cos \theta_A + \sin \theta_A$ and v_A -images $0, -\alpha_1 \sin \theta_A, -\alpha_1 \sin \theta_A + \cos \theta_A$; both triples are of the form $\{0, p, p+q\}$, so (1.1) applies to each and the two spans add up to

$$\begin{aligned} &\text{span}(u_A \cdot \Delta_{\alpha_1}) + \text{span}(v_A \cdot \Delta_{\alpha_1}) \\ &= \frac{1}{2}(|\alpha_1 \cos \theta_A| + |\alpha_1 \sin \theta_A| + |\cos \theta_A| + |\sin \theta_A|) \\ &\quad + \frac{1}{2}(|\alpha_1 \cos \theta_A + \sin \theta_A| + |-\alpha_1 \sin \theta_A + \cos \theta_A|) \\ &= \frac{1}{2}(|\alpha_1| \rho_A + \rho_A + \psi_A(\alpha_1)) \\ &= \frac{1}{2}(\psi_A(\alpha_1) + \rho_A(1 + |\alpha_1|)) = K_A. \end{aligned}$$

The six absolute values assemble exactly into $\rho_A, |\alpha_1| \rho_A$ and $\psi_A(\alpha_1)$. Each of these bounding boxes is a rectangle containing the corresponding triangle, so Lemmas 3 and 4 give

$$K_A \geq 1 + |\alpha_1|, \quad (3.7)$$

$$K_C \geq 1 + |\alpha_2|, \quad (3.8)$$

$$K_B \geq 1 + C_\alpha - |\alpha_1|, \quad (3.9)$$

$$K_B \geq 1 + C_\alpha - |\alpha_2|. \quad (3.10)$$

Step 5: pairwise lower bounds. Let $M = 2 + C_\alpha$. Then (3.7) + (3.9) and (3.8) + (3.10) give $K_A + K_B \geq M$ and $K_B + K_C \geq M$; and since $|\alpha_1| + |\alpha_2| \geq C_\alpha$, adding (3.7) and (3.8) gives $K_A + K_C \geq M$ as well. Adding (3.8) to $K_A + K_B \geq M$,

$$K_\Sigma := K_A + K_B + K_C \geq M + 1. \quad (3.11)$$

Step 6: conclusion of the proof. Put $t_i = 1 - s_i \geq 0$, so that $t_A + t_B + t_C = 3 - S < 3 - H$. From the pairwise bounds of Step 5, each $K_i \leq K_\Sigma - M$ (subtract the bound on the other two from K_Σ), and $K_\Sigma - M \geq 1 > 0$ by (3.11). Hence

$$\begin{aligned} \sum_i K_i s_i &= K_\Sigma - \sum_i K_i t_i \\ &\geq K_\Sigma - (K_\Sigma - M) \sum_i t_i \\ &> K_\Sigma - (K_\Sigma - M)(3 - H) \\ &= M + (H - 2)(K_\Sigma - M). \end{aligned}$$

Since $H - 2 \geq 0$ and $K_\Sigma - M \geq 1$, the last expression satisfies

$$M + (H - 2)(K_\Sigma - M) \geq M + (H - 2) = (2 + C_\alpha) + (H - 2) = H + C_\alpha.$$

Thus $\sum_i K_i s_i > H + C_\alpha$, contradicting (3.6) and completing the proof. $\square$

Remark 6 (Attainment). For every $H \in [2, 3]$ the value H is attained: three squares of sides $1, \frac{H-1}{2}, \frac{H-1}{2}$, stacked vertically, have total side length $1 + (H - 1) = H$.

Remark 7 (Sharpness of $H \geq 2$). The threshold is sharp throughout $1 \leq H < 2$. For $1 \leq H \leq \frac{3}{2}$, the three-square construction of sides $\frac{1}{2}, \frac{1}{2}, H - \frac{1}{2}$ has total side length $H + \frac{1}{2} > H$. For $\frac{3}{2} \leq H < 2$, a unit square together with two squares of side $\frac{1}{2}$, placed side by side above it, has total side length $2 > H$.

Remark 8 (Where each ingredient is used). Convexity gives the separating lines; the square shape is used through the support function ψ and through the fact that the projection of a square has length at least its side; the container is used through the wall inequalities. In the proof of the Strip Theorem, the hypothesis $H \geq 2$ is used only in Step 1, to force every pairwise sum above 1, and in Step 6, where $H - 2 \geq 0$. Steps 2–5, including the master inequality and the coefficient bounds, remain valid for every positive container height once a common vertical line is available. No angle or slope is restricted at any point.

4. THREE SQUARES IN A $1 \times x$ RECTANGLE

Let $G_3(x)$ denote the maximum total side length of three interior-disjoint squares, with arbitrary orientations, contained in a $1 \times x$ rectangle, where $x \geq 1$.

Theorem 9. *For every $x \geq 1$,*

$$G_3(x) = \begin{cases} x + \frac{1}{2}, & 1 \leq x \leq \frac{3}{2}, \\ 2, & \frac{3}{2} \leq x \leq 2, \\ x, & 2 \leq x \leq 3, \\ 3, & x \geq 3. \end{cases}$$

Proof. The upper bounds for $x \geq \frac{3}{2}$ are immediate from Theorem 1: for $\frac{3}{2} \leq x \leq 2$, enlarge the container to a 1×2 rectangle; for $2 \leq x \leq 3$, apply the theorem directly; and for $x \geq 3$, each side is at most 1.

It remains to prove

$$G_3(x) \leq x + \frac{1}{2} \quad (1 \leq x \leq \frac{3}{2}).$$

Suppose to the contrary that three squares have sides s_A, s_B, s_C and

$$S := s_A + s_B + s_C > x + \frac{1}{2}.$$

As in the proof of Theorem 1, shrink the squares slightly while preserving this strict inequality, so all separations may be taken strict. Any two of the squares lie in an $x \times x$ square, because $[0, 1] \times [0, x] \subseteq [0, x]^2$. Theorem 12, after scaling, therefore gives

$$s_i + s_j \leq x \quad (i \neq j). \quad (4.1)$$

If, say, $s_A + s_B \leq 1$, then adding this inequality to the two relevant instances of (4.1) gives

$$2S = (s_A + s_B) + (s_A + s_C) + (s_B + s_C) \leq 1 + 2x,$$

contrary to $S > x + \frac{1}{2}$. Hence every pairwise sum is greater than 1.

The horizontal projections consequently overlap pairwise in their interiors, so they have a common abscissa. Order the squares along the corresponding vertical line as A (bottom), B (middle), and C (top). Steps 2–4 of the proof of Theorem 1 apply verbatim with H replaced by x . Thus there are real numbers α_1, α_2 and coefficients K_A, K_B, K_C satisfying

$$K_A s_A + K_B s_B + K_C s_C \leq x + C_\alpha, \quad (4.2)$$

where

$$C_\alpha = \frac{1}{2}(|\alpha_1| + |\alpha_1 - \alpha_2| + |\alpha_2|),$$

and (3.7)–(3.10) hold. Put

$$L_A = K_A - 1, \quad L_B = K_B - 1, \quad L_C = K_C - 1.$$

We claim that

$$L_A s_A + L_B s_B + L_C s_C \geq C_\alpha - \frac{1}{2}. \quad (4.3)$$

Set $p = |\alpha_1|$ and $q = |\alpha_2|$. The estimates below are symmetric in (p, s_A) and (q, s_C) , so assume $p \geq q$.

If $\alpha_1 \alpha_2 \geq 0$, then $C_\alpha = p$, and (3.7)–(3.10) give

$$L_A \geq p, \quad L_B \geq p - q, \quad L_C \geq q.$$

Consequently,

$$L_A s_A + L_B s_B + L_C s_C \geq (p - q)(s_A + s_B) + q(s_A + s_C) \geq p = C_\alpha.$$

Suppose next that $\alpha_1 \alpha_2 < 0$. Then $C_\alpha = p + q$, and the same coefficient bounds give

$$L_A \geq p, \quad L_B \geq p, \quad L_C \geq q.$$

If $q \leq 1$, then

$$\begin{aligned} L_A s_A + L_B s_B + L_C s_C &\geq \left(p - \frac{q}{2}\right)(s_A + s_B) + \frac{q}{2}(s_A + s_C) + \frac{q}{2}(s_B + s_C) \\ &\geq p + \frac{q}{2} \geq p + q - \frac{1}{2} = C_\alpha - \frac{1}{2}. \end{aligned}$$

If $q > 1$, then after a reflection and, if necessary, interchanging p and q , the triangle $T(\alpha_1, \alpha_2)$ associated with K_B in Step 4 is congruent to the triangle in Lemma 5. Hence $K_B \geq C_\alpha + \frac{1}{2}$, or $L_B \geq C_\alpha - \frac{1}{2}$. Using also $s_B \leq 1$ and the strict pairwise inequalities,

$$\begin{aligned} L_A s_A + L_B s_B + L_C s_C &\geq p s_A + \left(C_\alpha - \frac{1}{2}\right) s_B + q s_C \\ &= p(s_A + s_B) + q(s_B + s_C) - \frac{1}{2} s_B \\ &> p + q - \frac{1}{2} s_B \geq C_\alpha - \frac{1}{2}. \end{aligned}$$

This proves (4.3). Therefore

$$K_A s_A + K_B s_B + K_C s_C = S + L_A s_A + L_B s_B + L_C s_C > x + C_\alpha,$$

contradicting (4.2).

The matching constructions are axis-parallel. For $1 \leq x \leq \frac{3}{2}$, use sides $\frac{1}{2}, \frac{1}{2}, x - \frac{1}{2}$; for $\frac{3}{2} \leq x \leq 2$, use sides $1, \frac{1}{2}, \frac{1}{2}$; for $2 \leq x \leq 3$, use sides $1, \frac{x-1}{2}, \frac{x-1}{2}$; and for $x \geq 3$, stack three unit squares. These attain the four displayed values. $\square$

This completes the $n = 3$ case of the rectangular square-packing question posed by Richard Stanley in a 2021 comment on [5].

By scaling and, if necessary, interchanging the coordinate axes, Theorem 9 determines the maximum total side length of three squares in every rectangle.

Corollary 10 (Rescaled strip form). *Let $F_3(u)$ denote the maximum total side length of three interior-disjoint squares in $[0, u] \times [0, 1]$, where $0 < u \leq 1$. Then*

$$F_3(u) = u G_3(1/u) = \begin{cases} 3u, & 0 < u \leq \frac{1}{3}, \\ 1, & \frac{1}{3} \leq u \leq \frac{1}{2}, \\ 2u, & \frac{1}{2} \leq u \leq \frac{2}{3}, \\ 1 + \frac{u}{2}, & \frac{2}{3} \leq u \leq 1. \end{cases}$$

In particular, $F_3(u) \leq 2u$ for every $\frac{1}{2} \leq u \leq 1$.

5. AXIS-PARALLEL GUILLOTINE CUTS IN FIVE-SQUARE PACKINGS

Corollary 11. *Let five interior-disjoint squares in the unit square have total side length S_5 . Suppose that the packing admits an axis-parallel guillotine cut, namely, an axis-parallel line that avoids the interiors of all five squares and separates them into two nonempty subfamilies. Then $S_5 \leq 2$.*

Proof. After rotating the entire configuration through a right angle if necessary, we may assume that ℓ is vertical, say $\ell = \{x = t\}$. Reflecting in the vertical midline if necessary, we may further assume that k squares lie on the left and $5 - k$ on the right, with $k \in \{1, 2\}$.

Case 1 + 4. The single left square, of side a , lies in $[0, t] \times [0, 1]$ and its horizontal projection has length at least a , so $t \geq a$. The other four squares lie in the right strip $[t, 1] \times [0, 1]$, of area $1 - t \leq 1 - a$, and have disjoint interiors, so their areas satisfy

$$\sum_{i=1}^4 b_i^2 \leq 1 - a.$$

By Cauchy–Schwarz, $(\sum b_i)^2 \leq 4 \sum b_i^2$, hence $\sum b_i \leq 2\sqrt{\sum b_i^2} \leq 2\sqrt{1-a}$ and

$$S_5 \leq a + 2\sqrt{1-a}.$$

Since $a \leq 1$ we have $2-a > 0$, so squaring is legitimate and

$$a + 2\sqrt{1-a} \leq 2 \iff 2\sqrt{1-a} \leq 2-a \iff 4(1-a) \leq (2-a)^2 \iff 0 \leq a^2,$$

which holds. Hence $S_5 \leq 2$.

Case 2 + 3. Let the right strip have width u , so the left one has width $1-u$. The two left squares contribute at most $\min(1, 2(1-u))$: each side is at most the strip width, and the two-square theorem (Theorem 12) in the ambient unit square caps the pair at 1. The three right squares, in a strip $u \times 1$, contribute at most $3u$ for $u \leq \frac{1}{3}$, at most 1 for $\frac{1}{3} \leq u \leq \frac{1}{2}$, and at most $2u$ for $u \geq \frac{1}{2}$, by Corollary 10. Then

$$\begin{aligned} u \leq \frac{1}{3} : \quad S_5 &\leq 1 + 3u \leq 2; \\ \frac{1}{3} \leq u \leq \frac{1}{2} : \quad S_5 &\leq 1 + 1 = 2; \\ u \geq \frac{1}{2} : \quad S_5 &\leq 2(1-u) + 2u = 2. \end{aligned} \quad \square$$

Consequence for Erdős problem #106. Consequently, any five-square packing in the unit square with total side length greater than 2 admits no axis-parallel guillotine cut. The equality $f(5) = 2$ is attributed to Newman through a personal communication to Erdős [4]; we are not aware of a published proof.

APPENDIX A. THE TWO-SQUARE THEOREM AND SMALL VALUES

Theorem 12. *Two squares with disjoint interiors in the unit square, with sides a and b , satisfy*

$$a + b \leq 1.$$

The result is attributed to Erdős in an early article in a Hungarian journal for secondary-school students; see the history notes at [4]. We give the following self-contained reconstruction.

Lemma 13 (corner depth). *A square Q of side s contained in the first quadrant contains a point z with $z_x \geq s$ and $z_y \geq s$; if Q is not axis-parallel, both inequalities are strict.*

Proof. Reducing the orientation modulo $\pi/2$, take $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$ and $\rho = \cos \theta + |\sin \theta| \in [1, \sqrt{2}]$. The horizontal and vertical half-widths of Q are $s\rho/2$, so its center c satisfies $c_x, c_y \geq s\rho/2$. Put $z = c + \frac{s}{2\rho}(1, 1)$. Then

$$|(z-c) \cdot u| = \frac{s}{2\rho} |\cos \theta + \sin \theta| \leq \frac{s}{2}, \quad |(z-c) \cdot v| = \frac{s}{2\rho} |\cos \theta - \sin \theta| \leq \frac{s}{2},$$

so $z \in Q$; and $z_x, z_y \geq \frac{s}{2}(\rho + \frac{1}{\rho}) \geq s$ by AM–GM, strictly if $\rho > 1$. $\square$

Proof of Theorem 12. The interiors are disjoint convex sets, so a line $px + qy = t$ separates the squares; composing with the reflections $x \mapsto 1-x$ and $y \mapsto 1-y$ of the container we may take $p, q \geq 0$ with $p+q > 0$, and the a -square in $px + qy \leq t$. Lemma 13 gives a point z of the a -square with $z_x, z_y \geq a$, whence $t \geq pz_x + qz_y \geq a(p+q)$. Reflecting the b -square through the center of the container and applying the lemma gives a point z' of the b -square with $z'_x, z'_y \leq 1-b$, whence $t \leq (1-b)(p+q)$. Dividing by $p+q$ gives $a+b \leq 1$. $\square$

Corollary 14. $f(3) = \frac{3}{2}$ and $f(4) = 2$.

Proof. For three squares of sides a, b, c , summing $a+b \leq 1$ over the three pairs gives $2(a+b+c) \leq 3$; three squares of side $\frac{1}{2}$ in an L attain it. For $f(4)$, Cauchy–Schwarz gives $\sum s_i \leq 2\sqrt{\sum s_i^2} \leq 2$, attained by the 2×2 grid. $\square$

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